

PW EARNERS
DSA in C++

Daily Practice Problems (DPP)

Topic: Basics of C++

Q1. State **True** or **False** for each statement below.

- (i) In C++, a `double` variable can store 3.14, while an `int` would truncate it to 3.
- (ii) The modulus operator `%` works with `double` operands just like with `int` operands.
- (iii) Blank spaces are allowed inside a variable name in C++ (e.g., `my variable` is valid).
- (iv) The post-increment operator (`x++`) first uses the current value in the expression, then increments it.

Q2. Identify whether each of the following is a **valid** or **invalid** C++ variable name. For every invalid name, write a brief reason.

Variable Name	Valid / Invalid	Reason (if invalid)
(i) <code>totalMarks</code>		
(ii) <code>_counter</code>		
(iii) <code>2ndPlayer</code>		
(iv) <code>player#score</code>		
(v) <code>int</code>		
(vi) <code>Hello World</code>		
(vii) <code>MAX_SIZE</code>		
(viii) <code>my, var</code>		
(ix) <code>__x1</code>		
(x) <code>Float</code>		

Q3. Choose the correct answer for each part.

- (i) What is the result of `17 % 5` in C++?

- A. 3 B. 2 C. 1 D. 0

(ii) Which declaration correctly stores 3.14159?

- A. `int pi = 3.14159;` B. `char pi = 3.14159;` C. `double pi = 3.14159;` D. `bool pi = 3.14159;`

(iii) If `int a = 5, b = 2;`, what does `a / b` give?

- A. 2.5 B. 2 C. 3 D. 2.0

For Q4 to Q10: Write the exact output of each program. If there is a compilation error, write "Compilation Error" and state why.

Q4.

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     int x = 10;
5     cout << x++ << endl;
6     cout << x << endl;
7     return 0;
8 }
```

Output:

.....

Q5.

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     int x = 10;
5     cout << ++x << endl;
6     cout << x << endl;
7     return 0;
8 }
```

Output:

.....

Q6.

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     int a = 7, b = 3;
5     cout << a % b << endl;
6     cout << (-7) % 3 << endl;
7     return 0;
8 }
```

Output:

.....

Q7.

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     int    a = 5;
5     double b = 2.0;
6     cout << a / 2 << endl;
7     cout << a / b << endl;
8     return 0;
9 }
```

Output:

.....

Q8.

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     char ch = 'A';
5     cout << ch << endl;
6     cout << (int)ch << endl;
7     cout << ch + 1 << endl;
8     return 0;
9 }
```

Output:

.....

Q9.

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     int x = 5;
5     int y = x++ + ++x;
6     cout << y << endl;
7     cout << x << endl;
8     return 0;
9 }
```

Output:

.....

Q10.

```
1 #include<iostream>
2 using namespace std;
3 int main(){
4     double x = 5.9;
```

```

5   int    y = (int)x;
6   cout << y          << endl;
7   cout << (int)3.99 << endl;
8   return 0;
9 }

```

Output:

.....

For Q11 to Q20: Write clean, compilable C++ programs. Use appropriate data types, *cin* for input, and *cout* for output.

Q11. Take an integer n as input. Print the value of $4n + 9$.

Sample — Input: 5 $\Rightarrow$ Output: 29

Q12. Take four integers a , b , c , d as input. Compute and print $(a + b) \times (c + d)$.

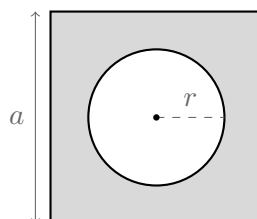
Sample — Input: 1 2 3 4 $\Rightarrow$ Output: 21

Q13. Write a program to calculate the **Volume of a Sphere** given radius r . Use $\pi = 3.14$.

$$V = \frac{4}{3} \pi r^3$$

Sample — Input: 3 $\Rightarrow$ Output: 113.04

Q14. A circle of radius r is inside a square of side a (the circle always sits completely inside the square without touching its edges). Find the area of the **shaded region** — the part inside the square but outside the circle. Use $\pi = 3.14$.



Sample — Input: $r = 2, a = 6 \Rightarrow$ Output: 23.44

Q15. Take a `char` as input and print its ASCII value.

Sample — Input: Z $\Rightarrow$ Output: 90

Q16. Take an integer `n` as input. Print the minimum non-negative integer that must be added to `n` to make it exactly divisible by 7. Use the modulus operator.

Sample — Input: 20 $\Rightarrow$ Output: 1 (since $20 + 1 = 21 = 7 \times 3$)

Q17. Take two integers `a` and `b` ($b \neq 0$) as input. Print the result of **ceiling integer division** of `a` by `b`.

Ceiling division always rounds up — e.g., $\lceil 7/2 \rceil = 4$, whereas normal integer division gives 3. Hint - Formula for positive integers: $(a + b - 1) / b$.

Sample — Input: 7 2 $\Rightarrow$ Output: 4

Q18. Compute the **Kinetic Energy** of a moving object. Take the Mass `m` (in kg, double) and Velocity `v` (in m/s, double) as input from the user. Be careful about integer vs. double division when applying the fraction coefficient.

$$\text{KE} = \frac{1}{2} \times m \times v^2$$

Sample — Input: `m = 10, v = 4` $\Rightarrow$ Output: 80

Q19. Take an amount in **Rupees** (double) and convert it to **Dollars**, given that **1 Dollar = 100 Rupees**.

Sample — Input: 250 $\Rightarrow$ Output: 2.5

Q20. A shopkeeper applies a discount of $d\%$ on the marked price m , and then charges a tax of $t\%$ on the discounted price. Take m , d , and t (all doubles) as input and print the final amount the customer pays.

$$\text{Discounted Price} = m - \frac{d}{100} \times m \quad \text{Final Price} = \text{Discounted Price} \times \left(1 + \frac{t}{100}\right)$$

Sample — Input: $m = 1000$, $d = 10$, $t = 18 \Rightarrow$ Output: 1062

All the best! Keep Coding ♡